FLEET TEAM DIGEST

737MAX-FTD-34-23009

Issue Title : Reports of Unexpected Vertical Deviation during VNAV Gradient Descents when using FMC Operational Flight Program Update 14.0

Airplane Model	ATA	Minor Model(s)
737MAX	3461-00	-10, -7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
	9E991	

Originated Date	Last Revised Date	Created On
12/21/2023	12/21/2023	10/18/2023
Estimated Completion Date	**Next Update Date**	

Description

Boeing has received reports of unexpected vertical deviation during descent from operators using U14.0 FMC operational flight program (OFP).

Previously, Fleet Team Digest article, 737MAX-FTD-34-23008 was issued by Boeing and provided information related to unexpected vertical deviation during descent when using the U13 FMC OFP. The FTD also advised the reported condition would be addressed in the U14.0 FMC OFP. However, after certification of the U14.0 FMC OFP, Boeing continues to receive reports of unexpected vertical deviation during descent associated with legs that have a coded vertical angle.

Boeing and our FMC supplier, GE Aviation, have identified additional conditions in the U14.0 FMC OFP that can result in unexpected vertical deviation occurrences during descent similar to the U13 FMC OFP occurrences. These occurrences have been replicated and software changes have been implemented to address them. These changes are included in the U14.1 FMC OFP. The attachment provides a profile view that includes a graphic vertical profile and describes the conditions that need to be present before the airplane can experience the unexpected vertical deviation event and also illustrates the differences in conditions for the U13 FMC OFP condition discussed in Boeing FTD 737MAX-FTD-34-23008.

The condition for the U14.0 VDEV issue is dissimilar to the U13 VDEV issue (discussed in 737MAX-FTD-34-23008) in that it requires a series of vertical angle legs with no intervening AT constraints; the U13 VDEV issue could occur with only a single vertical angle leg. It is also dissimilar to the U13 VDEV issue, in that it requires the last sequenced waypoint to have an AT constraint, whereas the U13 VDEV issue required an AT or ABOVE constraint. The conditions reported for U14.0 is similar in that a mis-compare is responsible for triggering a re-prediction. It is also similar in that it results in a path steeper than the coded vertical angle with the aircraft leveling-out to capture the steeper path. Both issues will always result in a steeper descent path. It may lead to a go-around, but it will never cause guidance below the intended descent path.

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Since the FMC OPC is common, this condition may also occur on the 737NG Fleet (reference 737NG-FTD-34-22003).

Status

The root cause for this condition has been addressed in the FMC OPS U14.1 software update (reference 737MAX-FTD-34-20002).

The FMC U14.1 OFP was certified mid November 2021 and incorporated into 737 MAX production airplanes at Airplane Line Position 8179 that was delivered November 2021. Service Bulletin 737-34-3656 was released to allow installation of the U14.1 FMC OFP in the 737 MAX fleet for line positions prior to 8179.

Interim Action

Operators are encouraged to review the attached illustration to familiarize themselves with this operation.

The recommendation for continued usage of the U14.0 FMC OFP is to remove and/or replace AT constraints on vertical angle legs with AT or ABOVE constraints where possible, or code AT constraints on all vertical angle legs in the FMC navigation database. The unexpected vertical deviation events when using the U14 FMC OFP can occur when there is a series of two or more vertical angle legs that use the same coded angle and there is no intervening AT constraint, and the last sequenced waypoint had an AT constraint. If a descent path re-computation is triggered in this condition, it will be incorrect and can lead to an unexpected vertical deviation.

We note that this revises the recommendations made in the interim action suggested in FTD 737MAX-FTD-34-23008 which recommended removing and/or replacing AT or ABOVE constraints on vertical angle legs with AT constraints.

Final Action

Incorporate FMC software U14.1 or later per Boeing's applicable service bulletin. SB issuance is noted in the milestones.

Be advise when operating FMC OPS U14.1 on airplanes equipped with the 2907C1 FMC hardware (Boeing P/N 10-62225-001/GE Aviation P/N 176200-01-01), there is an AFM Limitation for the 737 NG and MAX models, prohibiting RNP AR operations with minima below 0.3 nautical miles (NM)*. This limitation will be corrected with the issuance of the 2907C2 FMC hardware (Boeing P/N 10-62226-005, GE Aviation P/N 176200-02-01). Please refer to the FTD article, 737MAX-FTD-34-18008. *Operators flying 0.3 NM RNP minima and above; are not affected. No Airplane Flight Manual (AFM) Limitation is needed.

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Milestones

Service Bulletin Available: 737MAX: 28-May-2022 (Complete)

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737MAX-FTD-34-18008		
Fleet Team Digest Article (FTD)	737NG-FTD-34-22003		
Fleet Team Digest Article (FTD)	737MAX-FTD-34-20002		
Service Bulletin (SB)	737-34-3656		
Fleet Team Digest Article (FTD)	737MAX-FTD-34-23008		

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Flight Operations : Flight Operations
- New Equipment : Information on new parts available now or soon to be introduced

Attachments:

a) Unexpected Vertical Deviation

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